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Conception of an electronic instrument : The Theremin

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Contents

1 Introduction **2**

2 Pitch control circuit **2**
2.1 Frequency oscillator . . . 2
2.2 Low pass filter . . . 3
2.3 Variable frequency oscillator . . . 4
2.4 Mixer . . . 5
2.5 Envelope detector . . . 6

3 Audio output **7**
3.1 Inverting amplifier . . . 7
3.2 Power amplifier . . . 8

4 Seven segment controller **9**
4.1 Overall view . . . 9
4.2 Clock divider . . . 10
4.3 Refresh counter . . . 10
4.4 Anode control . . . 10
4.5 BCD control . . . 10
4.6 BCD to cathodes . . . 11

5 Binary to BCD converter **11**

6 8-bit counter on a seven segment display **11**
6.1 8-bit counter . . . 12

Bibliography **13**

1 Introduction

A theremin is an electronical instrument that is played without any physical contact between the thereminist and the theremin. This instrument is made of two antennas and an electronic circuit. The hand of the thereminist and the antenna act as two plates of a capacitor. When the thereminist's hand moves closer or further away from the antenna, the capacitor's value will then be modified. The purpose of the two antennas is to change the pitch and the volume of the instrument. In other words, the first antenna will modify the frequency of the output signal and the other one will change its amplitude.

The electronic circuit in the theremin is described in a block diagram shown in Fig 1. The theremin we built is mostly based on the different components of this diagram and is inspired from [1] and [2].

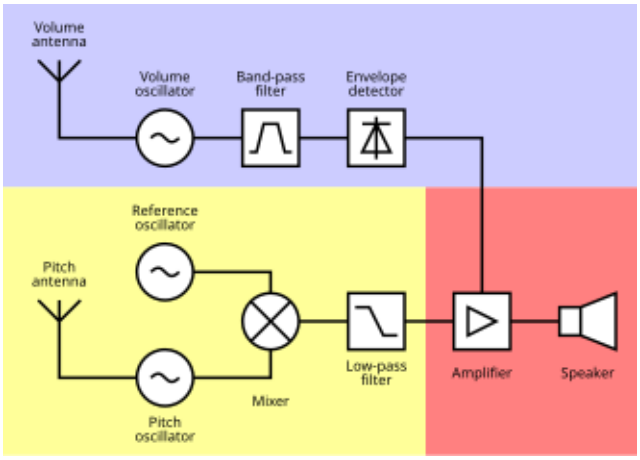


Figure 1: Block diagram of a theremin. In blue is the volume control circuit, in yellow is the pitch control circuit and the audio output is in red [3].

2 Pitch control circuit

2.1 Frequency oscillator

As shown on Fig 1, the pitch antenna signal is mixed with some reference oscillator. The reference oscillator in our theremin is made with an OP 37 amplifier. This type of oscillator will produce a square output going from V_{sat} to $-V_{sat}$ signal without any input signal. Let's suppose that the output of the oscillator shown in Fig 2 is in positive saturation. The capacitor C_4 will then be charged exponentially up to the saturation potential due to the resistor R_4. The maximal potential that the capacitor can be charged up to never reaches this saturation potential because the voltage crosses the upper trip point (UTP). When this point is crossed, the output signal of the oscillator then switches to $-V_{sat}$. The output is then in negative saturation and the capacitor will discharge towards $-V_{sat}$. But once again before reaching this saturation value, the capacitor voltage will cross the lower trip point (LTP) which will switch the value of the ouput voltage to V_{sat}. This cycle will then be repeated periodically [4].

We built this type of oscillator with a capacitor $C_4 = 1$ nF and three resistors where $R_4 = 100$ kΩ $= R_5 = R_6$. The period of the output signal is given by :

$$T = 2R_4\, C_4\, \ln\left(\frac{1+B}{1-B}\right), \tag{1}$$

where B is the feedback fraction :

$$B = \frac{R_6}{R_5 + R_6}. \tag{2}$$

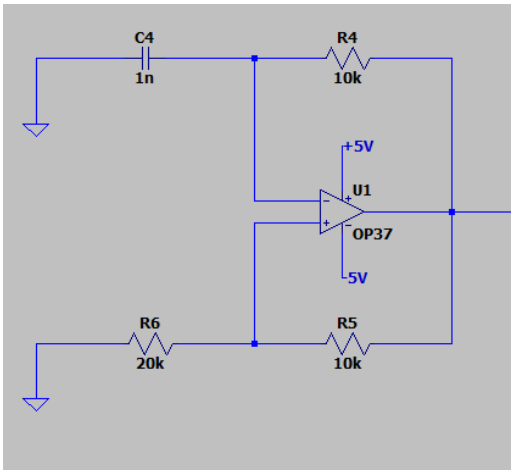


Figure 2: Schematics of the fixed frequency oscillator

For our fixed frequency oscillator, the feedback fraction has a value of $B = 0,666$ and the period is $T = 32,19\ \mu$s. The frequency of the output signal can then be computed :

$$f_{fix} = \frac{1}{T} = 31,07 \text{ kHz}.$$

This value matches the experimental frequency of the oscillator, which was measured to be $f_{mes} = 32,61$ kHz. The signal obtained at the output of the oscillator is shown in Fig 3. The shape of the signal is close to a rectangular wave. This inconsistency is probably due to the type of amplifier used in the circuit.

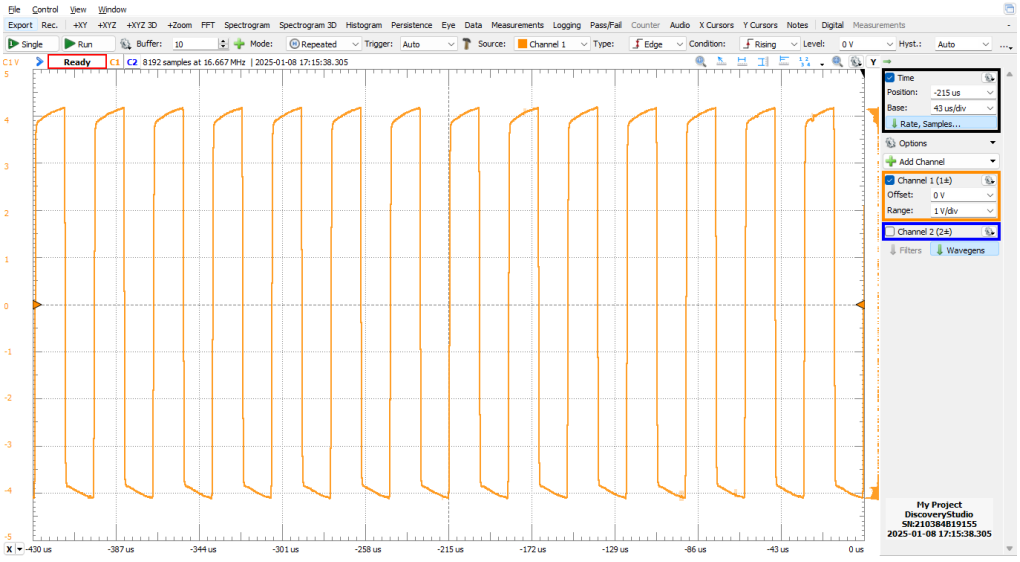


Figure 3: Signal at the output of the fixed frequency oscillator.

2.2 Low pass filter

Before going into the mixer, the signal of the fixed frequency oscillator goes into a low pass filter. This filter will reduce the noise generated in the oscillator. This filter is shown in Fig 4 and it is composed of three

low pass filters with the same cutoff frequency. Having three filters in series will greatly improve the higher frequency suppression and it also makes the filter more selective.

The cutoff frequency of each of these filters is :

$$f_{LP_1} = \frac{1}{2\pi RC} = 397,887 \text{ kHz}.$$

The signal at the output of the low pass filter will be modified to a triangular signal as can be seen in Fig 5. This signal will be heterodyned with the signal from the variable frequency oscillator so in the end, this triangular shaped signal will work fine.

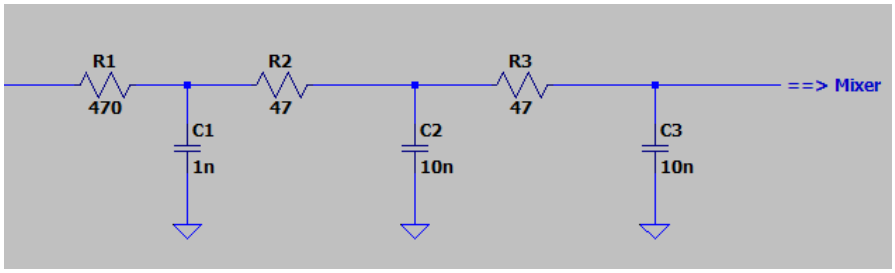


Figure 4: Schematics of the low pass filter.

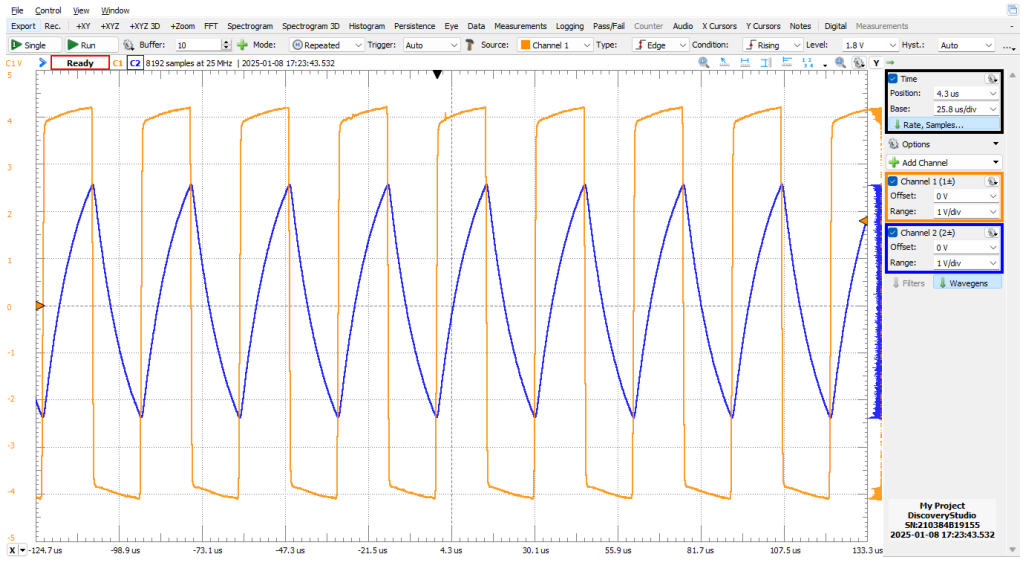


Figure 5: Signal from the low pass filter. The input, from the fixed frequency oscillator, is in yellow and the ouptut is in blue.

2.3 Variable frequency oscillator

The pitch antenna is connected to another relaxation oscillator. This oscillator is made of an OP27 amplifier, a 100 pF capacitor and three 100 kΩ resistors and a potentiometer as can be seen in Fig 6. The antenna is connected in parallel of the capacitor which will increase the value of the equivalent capacity. For capacitors in parallel, the equivalent capacity is given by

$$C_{eq} = \Sigma\, C_i.$$

By moving our hand in regards to the antenna, the equivalent capacity will be modified and this will modify the RC time constant in Equation 1. This modification in the period is inversely proportional to the modification

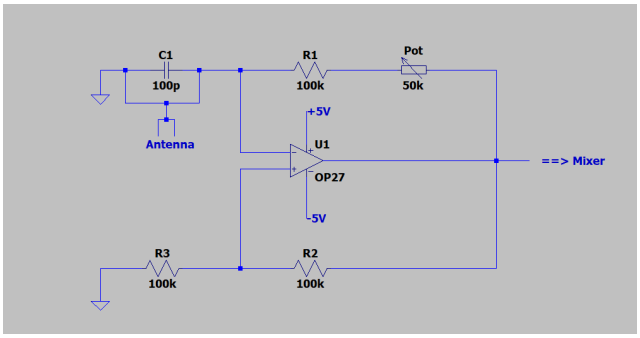


Figure 6: Schematics of the variable frequency oscillator.

of the oscillator frequency. The potentiometer in series with resistance R_1 allows to chose an output frequency. Adding this potentiometer will offer a certain freedom in the tuning of the output pitch of the theremin.

The frequency of the oscillator when the antenna is not connected is calculated using Equations 1 and 2:

- $f_{var_0} = 45,51$ kHz when the potentiometer is set to 0 Ω;
- $f_{var_{50}} = 30,34$ kHz when the potentiometer is set to 50 kΩ.

The signal from this variable frequency oscillator is shown in Fig 7, when the antenna is disconnected and the potentiometer is set to 0 Ω. The measured frequency of this oscillator is $f_{mes} = 34,37$ kHz. This frequency is different than the calculated frequency for the oscillator but it still corresponds to a working frequency for a theremin. The shape of the signal is different than the one from the reference oscillator because the two oscillators are made of two different operational amplifiers.

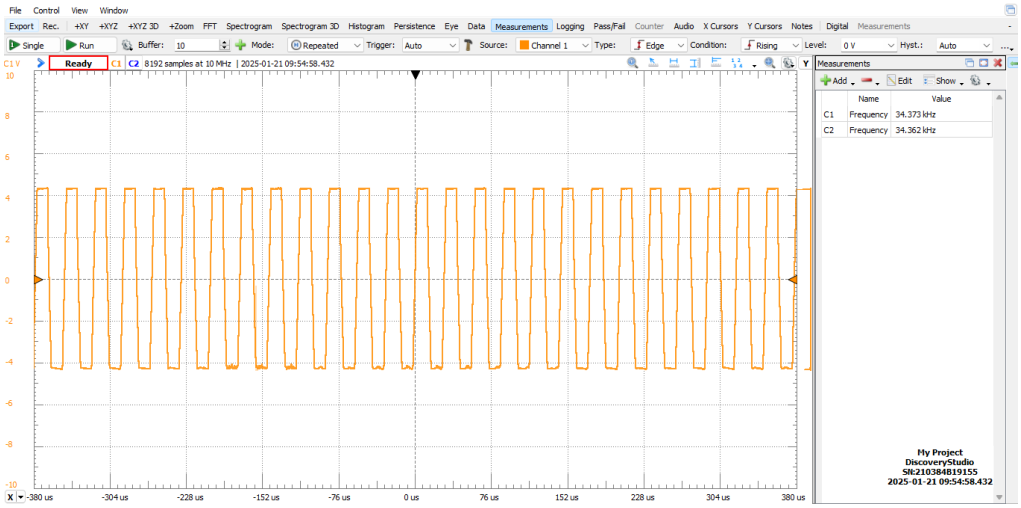


Figure 7: Signal at the output of the variable frequency oscillator. This signal was obtained when the antenna was not connected to the circuit.

2.4 Mixer

The signals coming from both oscillators are mixed together using the mixer shown in Fig 8. This circuit is composed of an OP27 amplifier, two resistors of 47 kΩ and one feedback resistor of 20 kΩ. This mixer will subtract the incoming signals. The output signal from this mixer is a beating as shown in Fig 9. The frequency

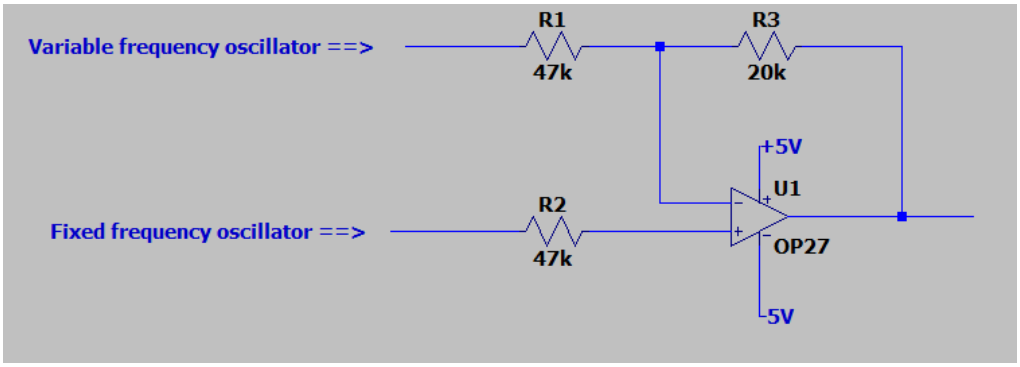


Figure 8: Schematics of the mixer.

of this beating is $f_{beat} = 32,70$ kHz. This frequency is outside of the audio range frequency, 20 to 20 kHz, so this signal can not yet be used as the output signal of the theremin.

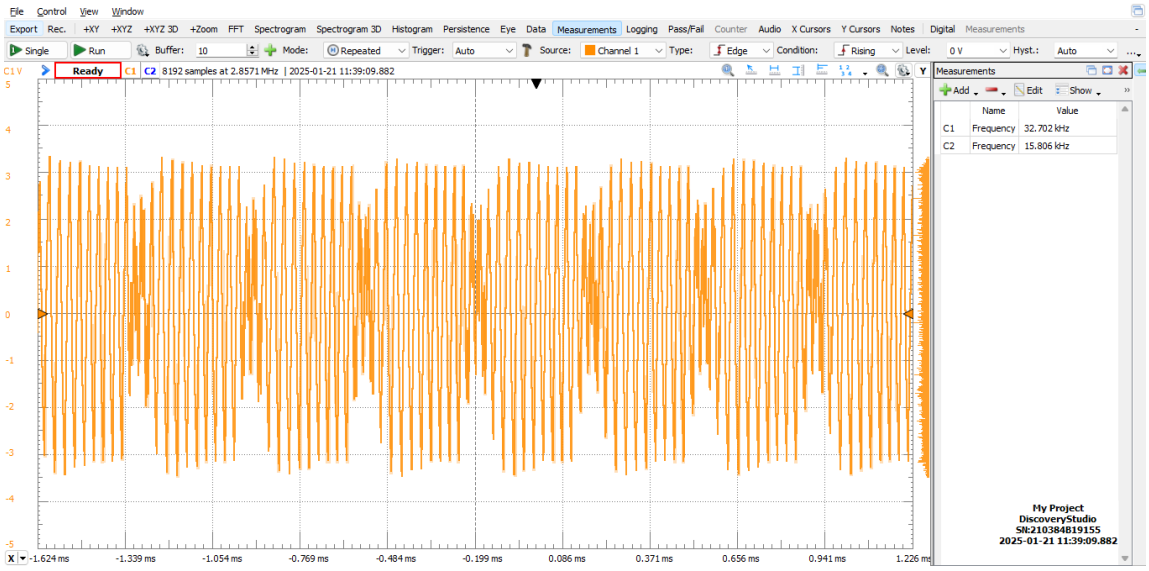


Figure 9: Beating signal at the output of the mixer.

2.5 Envelope detector

Because the beat frequency is too high to be heard, a low pass filter and an envelope detector is placed after the mixer. The low pass filter has a cutoff frequency of

$$f_{LP_2} = \frac{1}{2\pi RC} = 338,6 \text{ kHz},$$

which will reduce the unwanted noise that might have been created after the mixer.

The envelope detector is made of a diode, a resistor and a capacitor as shown in Fig 10. In this type of circuit, the time constant RC must be chosen to be greater than the period of the input signal [4]. For this envelope detector, $RC = 10 \cdot 10^3 \cdot 0,047 \cdot 10^{-6} = 470$ μs, which is bigger than the period of the mixer signal : $T = 1/f_{beat} = 29,1$ μs.

The output signal of the envelope detector is shown in Fig 11. This signal has a frequency of $f_{env} = 2,724$ kHz, which is in the audio frequency range. This signal can then be used to generated an output sound.

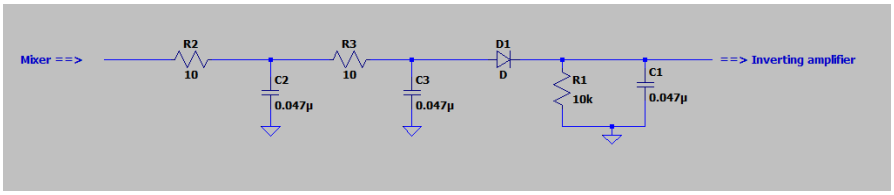


Figure 10: Schematics of the envelope detector.

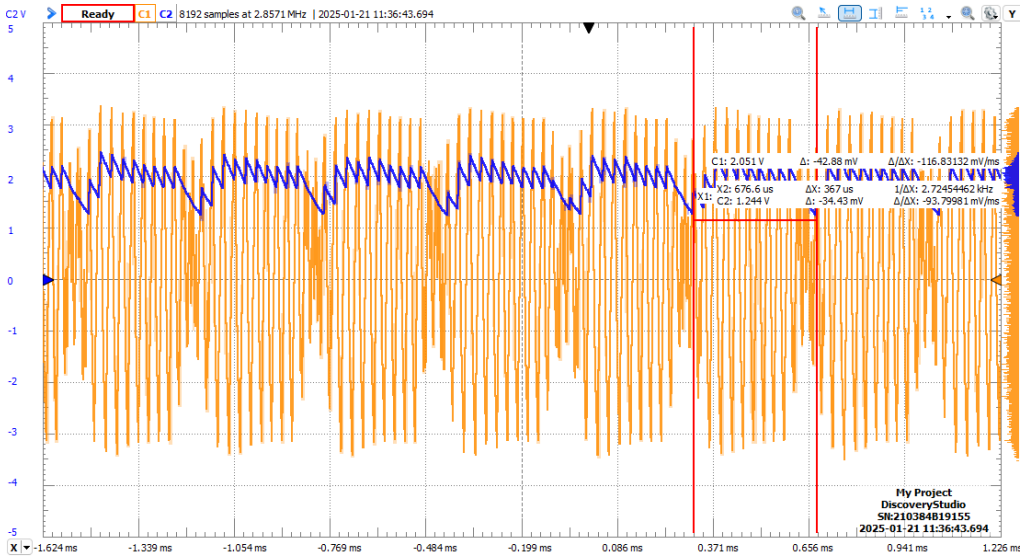


Figure 11: Envelope detector signal. The input signal from the mixer is in yellow and the ouptut signal is in blue.

3 Audio output

3.1 Inverting amplifier

The envelope detector signal is then sent into an inverting amplifier as shown in Fig 12. This amplifier is a simplified version of a voltage controlled amplifier. In this type of circuit, the gain is given by :

$$G = -\frac{R_f}{R_i}. \tag{3}$$

In our case, the feedback resistor R_f is a 10 kΩ potentiometer and the input resistor R_i is a 4,7 kΩ resistor.

The potentiometer allows to modify the value of the feedback resistor which will consequently modify the gain of the amplifier. Modifying the gain will then lead to an increase or a decrease of the amplitude of the signal and consequently in the soun coming out of the speaker.

The positive input of the OP37 is connected to the ground using a 5 MΩ resistor, which is the biggest resistor we can have. There is a capacitor at the output of this amplifier to decouple the output signal from the input of the power amplifier. The output signal of this inverting amplifier is shown in Fig 13. As can be seen on the figure, this signal is completely negative, which makes it not yet usable. If we connect the speaker at this output, no sound will come out of it because of this completely negative signal.

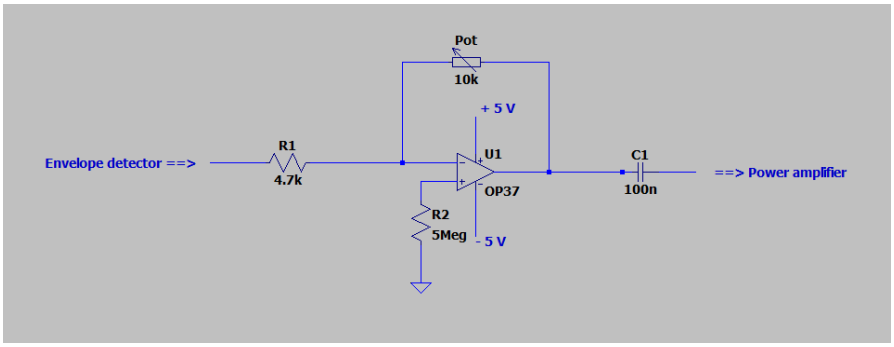


Figure 12: Schematics of the inverting amplifier.

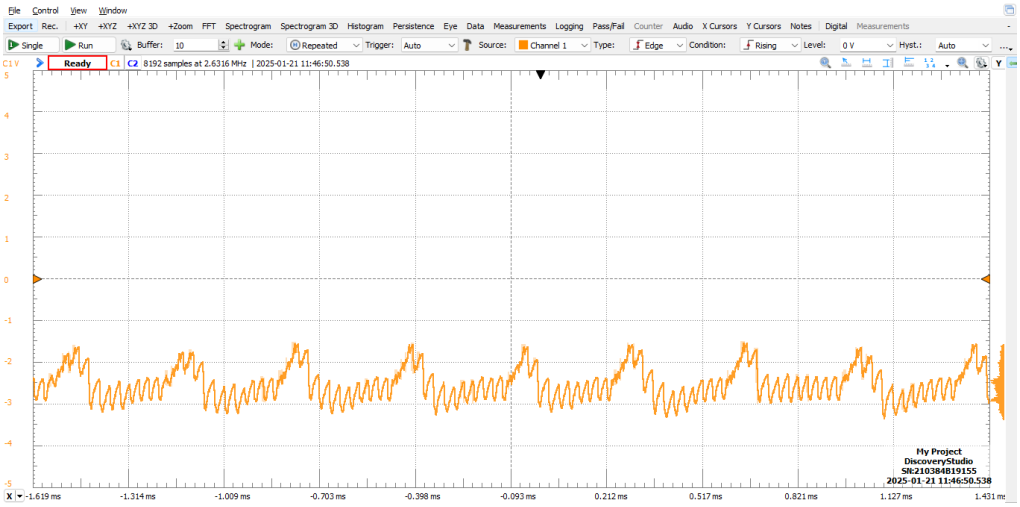


Figure 13: Output signals of the inverting amplifier. This signal was measured for a high R_f which means that the volume of sound coming from the speaker at the end was high.

3.2 Power amplifier

A diode biased class B push pull amplifier is used to provide the necessary current needed to increase the loudness of the speaker. This type of amplifier is necessary because the output current of the op-amp must be kept as low as possible (below ±10 mA according to the datasheet) and the 8 Ω speaker would draw more than 100 mA current at a 1V amplitude signal. The amplifier is shown in Fig 14. A class B amplifier is an amplifier in which each transistor in this amplifier clips off half of the ac cycle. The push-pull configuration is a configuration with two transistors, where one transistor conducts for one half of the ac cycle while the other does not and inversely. In our case, one transistor is an npn transistor and the other one is a pnp transistor. Both of these transistor are in the emitter follower configuration. And lastly, it is called diode biased because the two diodes will produce a bias voltage for the emitter of both transistors. These diodes are placed there to avoid thermal runaway, which is the increase of the collector voltage due to a thermal increase in the transistor [4].

The bias current through the diodes is :

$$I_{bias} = \frac{V_{CC} - 2V_{BE}}{2R} = \frac{5 - 1,4}{2 \cdot 10 \cdot 10^3} = 180\ \mu\text{A} \tag{4}$$

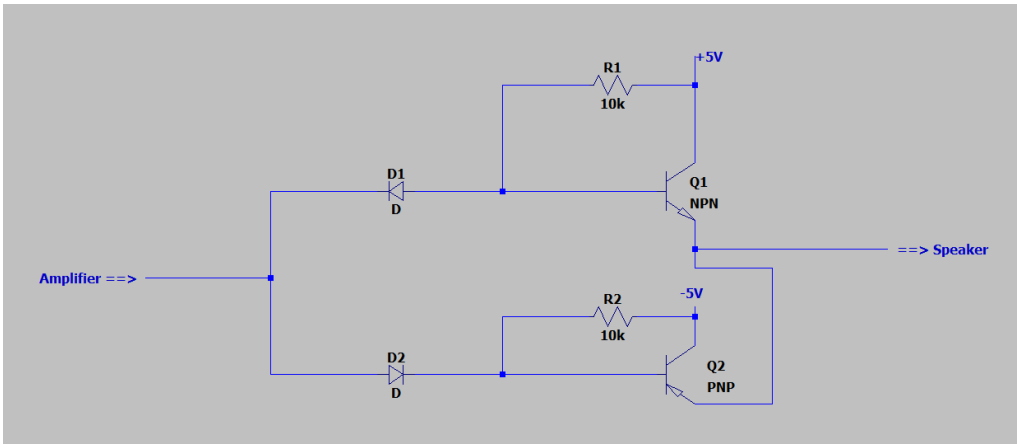


Figure 14: Schematics of the power amplifier

The collector saturation current is :

$$I_{C(sat)} = \frac{V_{CEQ}}{R_L} = \frac{V_{CC}}{2R_L} \tag{5}$$

In our circuit, the load resistor is the resistor of the speaker (8 Ω). This gives a saturation current of : $I_{C(sat)} = 0,312$ A. The total current drain is given by :

$$I_{dc} = I_{bias} + \frac{I_{C(sat)}}{\pi} = 180 \cdot 10^{-6} + 99,5 \cdot 10^{-3} = 99,6 \text{ mA} \tag{6}$$

The dc input power is : $P_{dc} = V_{CC} I_{dc} = 0,498$ W and the maximum ac output power is :

$$P_{out(max)} = \frac{MPP^2}{8R_L} = \frac{5^2}{64} = 0,391 \text{ W}. \tag{7}$$

The efficiency of our power amplifier is then : $\eta = P_{dc}/P_{out(max)} \cdot 100 = 128\%$, which will indeed increase the sound coming out of the speaker.

4 Seven segment controller

4.1 Overall view

Here is the entire module we used to code a seven segment controller in Vivado [5] that is represented in Fig 15. This module allows us to display different numbers from zero to nine on four of the anodes of the seven segment display. It is an essential part of the block design used to build our final timer which will be useful to measure for how long we vary the frequency of the signal by moving our hand near the pitch antenna. We will go through the different parts of this module in the following sections.

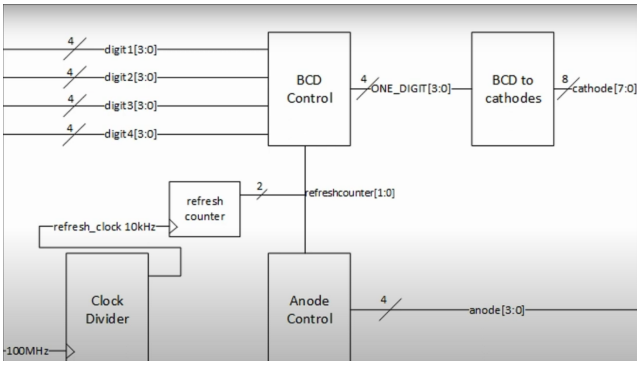


Figure 15: Seven segment module.

4.2 Clock divider

Firstly we code a clock divider module in order to slow down the frequency of the board's clock to go from 100 Mhz to 10 kHz. We can do that by implementing a counter which starts counting at each rising edge of the clock and counts up to division_value which we define using the formula :

$$\text{division_value} = \frac{\text{board's clock frequency}}{2 \cdot \text{desired frequency}} - 1 = 4999$$

With the desired frequency being 10 kHz and the board's clock frequency being 100 MHz in this case. The divided clock's value is 0 until the time the counter reaches division_value. At that time, we flip the clock signal and reset the counter. The divided clock's value is then 1. We count again up to division_value and flip the divided clock's value, which is again 0, and we obtain one pulse of the divided clock signal. So by increasing the variable division_value, the frequency of the divided clock gets smaller because we count for a longer time. This module is essential to see clearly the different numbers on the seven segment display because if the refresh clock is too fast then the numbers will blend in [6].

4.3 Refresh counter

This module is used to create a counter that counts from 0 to 3. The output of this counter is sent to the BCD control and anode control. It's important to note that putting the refresh clock at 10 kHz is still fast enough to trick the human eye into thinking that the digits are on at the same time, even though the digits are displayed one at a time based on the refresh counter module. This module is useful to make sure that when the refresh counter is for instance 00, the corresponding anode for digit 1 is turned on and the corresponding BCD value is passed through at the same time.

4.4 Anode control

This module choose which anode will be turned on. Indeed, we code that every time the refresh counter has an output 00, then anode one (the leftmost on the board) is on or similarly if it has an output 01 then anode two is on and it follows the same logic for 10 and 11.

4.5 BCD control

The basic principle behind BCD is that each decimal digit can be represented by a fixed number of bits. In this case, we use four bits to represent each one of the four decimal number at the input of the BCD control. For example if we want to display 125 on the seven segment display, each digit must be represented in binary and are then passed through the BCD control module separately. This module is used to pass BCD value for a corresponding value of the refresh counter output so that only one of the input BCD value is chosen (which we

name "ONE_DIGIT") and then pass through the BCD to cathode module. In fact, we code that every time the refresh counter has an output 00 then we let through "digit1" or similarly if it has an output 01 then let through "digit2" and it follows the same logic for 10 and 11. It is important to assign it this way so that it matches the output of the anode module and we see the digits composing the number in the right order.

4.6 BCD to cathodes

This module is used to turn the "ONE_DIGIT" BCD value into a decimal number on the seven segment display by turning on specific cathodes. All we did here is looked into the Nexys A7™ FPGA Board Reference Manual in order to turn on the right cathodes to display a specific decimal number.

5 Binary to BCD converter

In order to make our timer work, we will need to send 4 bit BCD values to the seven segment display which means that a new module is necessary in order to convert 8 bit binary values coming from the 8-bit counter to 4 bit BCD values. To code this converter, we first put the 8 bit binary value into a 20 bit wide shift register where the rest of the bits are initially 0 and are divided equally into 3 subregisters named one's, ten's and hundred's containing each 4 bit. Then we use the following algorithm : "is any of the one's, ten's and hundred's ≥ 5 in the beginning of the iteration. If yes, we add 3 and shift the values of the shift register left. If no, we just shift left [7]. Note that we use registers to keep track of the number of iteration. We stop the iterations once the initial 8 bit value is 0 and we obtain a 4 bit value in each of the 3 subregisters. At the end, we see that we have broken down the initial 8 bit binary value into one's, ten's and hundred's. In our code, we use 3 temporary registers in order to store the output value of one's, ten's and hundred's of the previous iteration until the final BCD value is ready.

6 8-bit counter on a seven segment display

In this section we are going to put together all the different modules that we have seen in order to construct our timer, as can be seen in Fig 16.

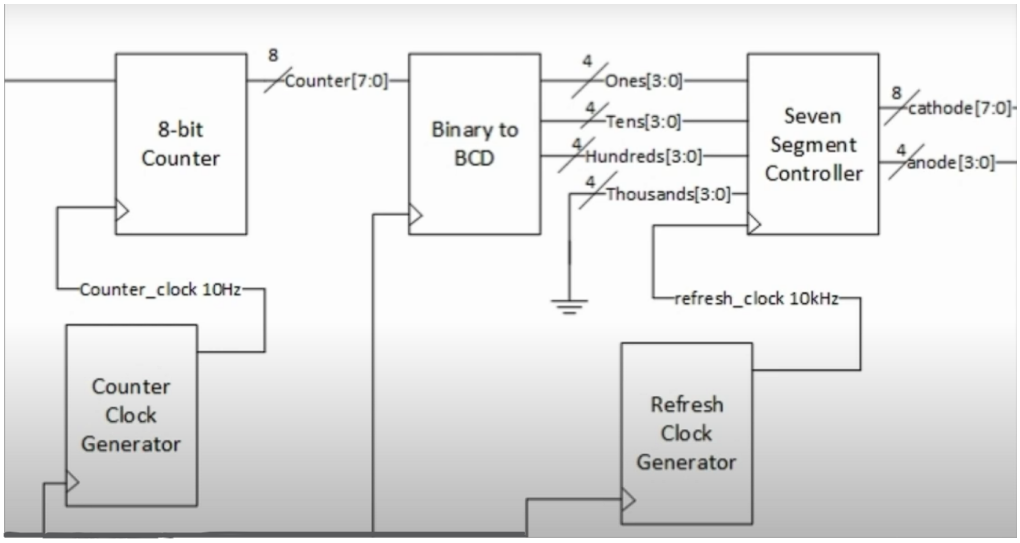


Figure 16: 8-bit counter on a seven segment display [8]

6.1 8-bit counter

In this block design we see three of the modules that we have seen before, which are the binary to BCD module, the seven segment controller and the clock divider. In this case, we use two clock dividers which still have a 100 MHz clock input. One of them is a refresh clock at a frequency of 10 kHz which handle the seven segment controller as before and the other is a counter clock which takes care of our new module : the 8-bit counter. We set the frequency of this clock to 1 Hz specifically because we want to count at a rate of one count per second for our timer to display the right time interval corresponding to a second. To set this clock, we can just modify a bit the module of the previous clock divider by choosing a division_value of 49999999 instead of 4999. The 8 bit counter module is a way to count to numbers up to 10000000 which correspond to 256 in decimal unit by using binary addition. So the maximum number the seven segment controller will be able to display is 255. Note that we set the output "Thousands" of the binary to BCD module to 0 because the limitations of the counter does not allow us to get a number of 4 digits at the input of the binary to BCD module. To initiate and reset the counter we simply use two of the switch assigned to our board as inputs of the 8 bit counter.

Bibliography

[1] GreatScottLab, "Make your own simple theremin," 2015, accessed: 2025-01-13. [Online]. Available: https://www.instructables.com/Make-Your-Own-Simple-Theremin/

[2] YouTube, "Theremin: The instrument that you play without touching," 2025, accessed: 2025-01-13. [Online]. Available: https://www.youtube.com/watch?v=97wjKQ-dKHo&t=10s

[3] Wikipedia contributors, "Theremin," 2025, accessed: 2025-01-13. [Online]. Available: https://en.wikipedia.org/wiki/Theremin

[4] A. Malvino and D. J. Bates, *Electronics Principles*, 7th ed. New York, NY: McGraw-Hill Education, 2007.

[5] GeeksforGeeks, "Getting started with verilog," https://www.geeksforgeeks.org/getting-started-with-verilog/, accessed: 2025-01-15.

[6] YouTube, "Video title (counters as timers)," https://www.youtube.com/watch?v=NqEfGWPfW78&t=285s, accessed: 2025-01-15.

[7] T. L. Floyd, *Digital Fundamentals*, 11th ed. Pearson, 2014.

[8] Gajalik, "Simple tutorials," https://gajalik.wixsite.com/simpletutorials, accessed: 2025-01-15.